Fourier Series

Fourier series are infinite series that represent periodic functions in terms of cosines and sines

A function f(x) is called a periodic function if f(x) is defined for all real x, except possibly at some points, and if there is some positive number p, called a period of f(x), such that

$$f(x+p) = f(x) \tag{1}$$

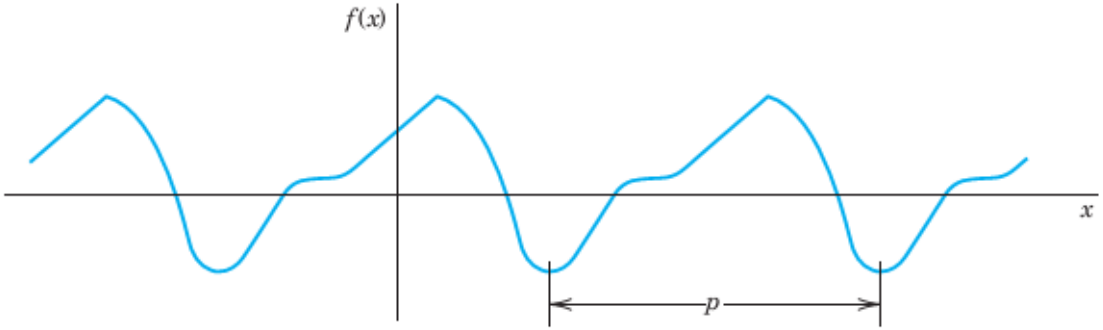


Fig. 258. Periodic function of period p

Fourier series

f(x) is a given function of period 2π:
→the Fourier series of f(x)

(5) $$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

where:

(6)

(0) $$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

(a) $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx \qquad n = 1, 2, \cdots$$

(b) $$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx \qquad n = 1, 2, \cdots$$

EXAMPLE

Periodic Rectangular Wave (Fig. 260)

Find the Fourier coefficients of the periodic function $f(x)$ in Fig. 260. The formula is

$$f(x) = \begin{cases} -k & \text{if} \quad -\pi < x < 0 \\ k & \text{if} \quad 0 < x < \pi \end{cases} \qquad \text{and} \qquad f(x + 2\pi) = f(x). \tag{7}$$

Functions of this kind occur as external forces acting on mechanical systems, electromotive forces in electric circuits, etc. (The value of $f(x)$ at a single point does not affect the integral; hence we can leave $f(x)$ undefined at $x = 0$ and $x = \pm\pi$.)

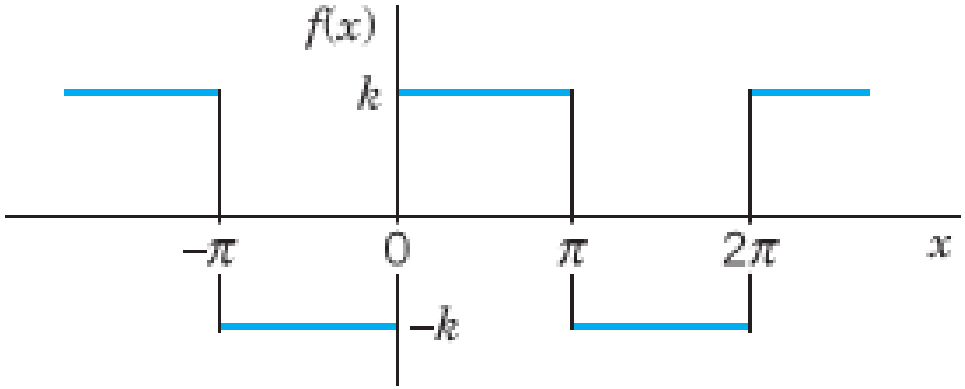


Fig. 260. Given function $f(x)$ (Periodic reactangular wave)

Page 477

$$f(x) = \begin{cases} -k & \text{if} \quad -\pi < x < 0 \\ k & \text{if} \quad 0 < x < \pi \end{cases} \qquad \text{and} \qquad f(x + 2\pi) = f(x).$$

Solution: a_0=0

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\cos nx\,dx + \int_{0}^{\pi} k\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-k\frac{\sin nx}{n}\Bigg|_{-\pi}^{0} + k\frac{\sin nx}{n}\Bigg|_{0}^{\pi}\right] = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-k)\sin nx\,dx + \int_{0}^{\pi} k\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\left[k\frac{\cos nx}{n}\Bigg|_{-\pi}^{0} - k\frac{\cos nx}{n}\Bigg|_{0}^{\pi}\right].$$

$$b_n = \frac{k}{n\pi}[\cos 0 - \cos(-n\pi) - \cos n\pi + \cos 0] = \frac{2k}{n\pi}(1 - \cos n\pi).$$

$$f(x) = \begin{cases} -k & \text{if} \quad -\pi < x < 0 \\ k & \text{if} \quad 0 < x < \pi \end{cases} \qquad \text{and} \qquad f(x + 2\pi) = f(x).$$

the Fourier series of $f(x)$ is

$$\frac{4k}{\pi}\left(\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \cdots\right).$$

- Một số công thức tích phân thường gặp

$$\int t\cos at\,dt = \frac{1}{a^2}\cos at + \frac{1}{a}t\sin at$$

$$\int t\sin at\,dt = \frac{1}{a^2}\sin at - \frac{1}{a}t\cos at$$

(9)

(a) $$\int_{-\pi}^{\pi} \cos nx \cos mx\,dx = 0 \qquad (n \neq m)$$

(b) $$\int_{-\pi}^{\pi} \sin nx \sin mx\,dx = 0 \qquad (n \neq m)$$

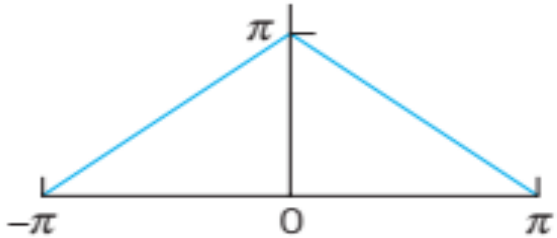
(c) $$\int_{-\pi}^{\pi} \sin nx \cos mx\,dx = 0 \qquad (n \neq m \text{ or } n = m).$$

PROBLEM SET 11.1

Find the Fourier series of the given function $f(x)$

9. $f(x) = \begin{cases} x & \text{if} \quad -\pi < x < 0 \\ \pi - x & \text{if} \quad 0 < x < \pi \end{cases}$

17.



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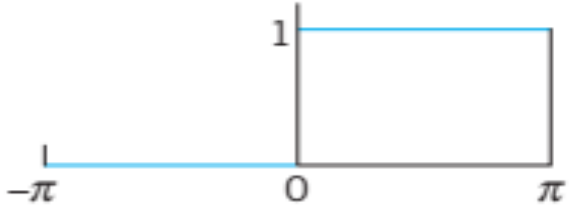

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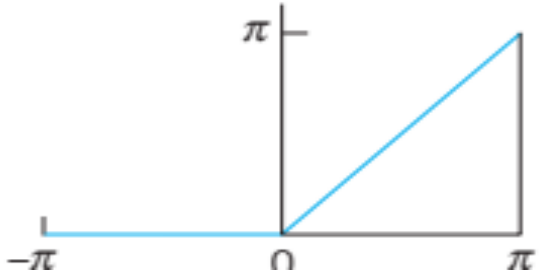

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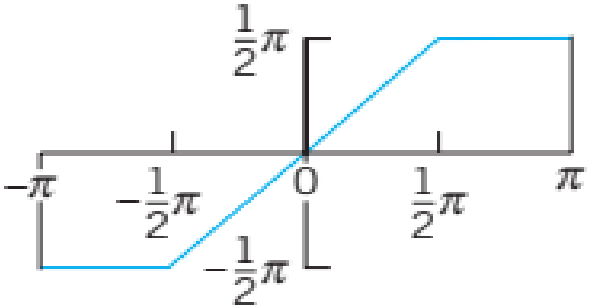

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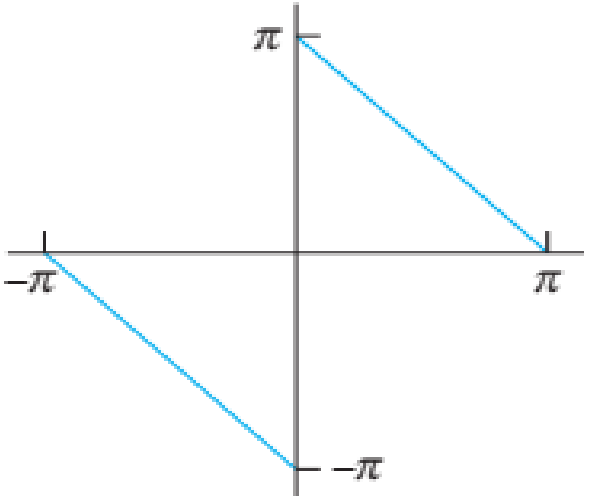


Fourier series

Fourier series of a function f(x) of period 2L:

(5)
$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi}{L}x + b_n \sin\frac{n\pi}{L}x\right)$$

(6)

(0)
$$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

(a)
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx \qquad n = 1, 2, \cdots$$

(b)
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx \qquad n = 1, 2, \cdots.$$

Periodic Rectangular Wave

Find the Fourier series of the function (Fig. 263)

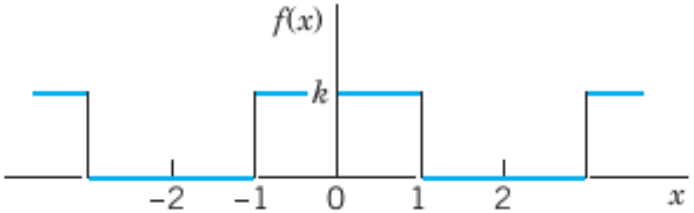


Fig. 263. Example 1

$$f(x) = \begin{cases} 0 & \text{if} \quad -2 < x < -1 \\ k & \text{if} \quad -1 < x < \quad 1 \\ 0 & \text{if} \quad \quad 1 < x < \quad 2 \end{cases} \qquad p = 2L = 4, \quad L = 2.$$

Solution. From (6.0) we obtain $a_0 = k/2$ (verify!). From (6a) we obtain

$$a_n = \frac{1}{2}\int_{-2}^{2} f(x)\cos\frac{n\pi x}{2}\,dx = \frac{1}{2}\int_{-1}^{1} k\cos\frac{n\pi x}{2}\,dx = \frac{2k}{n\pi}\sin\frac{n\pi}{2}.$$

Thus $a_n = 0$ if n is even and

$$a_n = 2k/n\pi \quad \text{if} \quad n = 1, 5, 9, \cdots, \qquad a_n = -2k/n\pi \quad \text{if} \quad n = 3, 7, 11, \cdots.$$

From (6b) we find that $b_n = 0$ for $n = 1, 2, \cdots$. Hence the Fourier series is a **Fourier cosine series** (that is, it has no sine terms)

$$f(x) = \frac{k}{2} + \frac{2k}{\pi}\left(\cos\frac{\pi}{2}x - \frac{1}{3}\cos\frac{3\pi}{2}x + \frac{1}{5}\cos\frac{5\pi}{2}x - + \cdots\right).$$

Simplifications: Even and Odd Functions

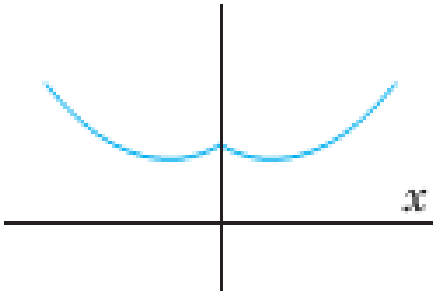
(5*) $$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi}{L} x \qquad (f \text{ even})$$

(6*) $$a_0 = \frac{1}{L}\int_0^L f(x)\,dx, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx, \qquad n = 1, 2, \cdots.$$

$$\omega_0 = \frac{2\pi}{T} = \frac{\pi}{L}\ (rad/s)$$

$$a_0 = \frac{2}{T}\int_0^{T/2} f(t)\,dt$$

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega_0 t\,dt$$



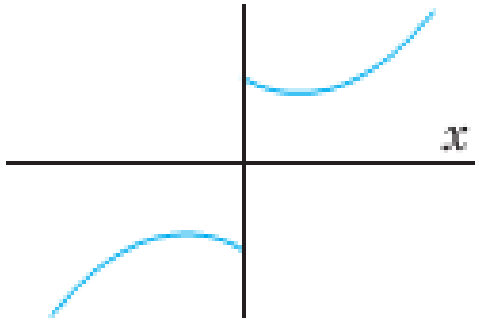
- Đồ thị đối xứng qua trục tung

(5)**	$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{L} x$	(f odd)

$$b_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

$$\omega_0 = \frac{2\pi}{T} = \frac{\pi}{L} \ (rad/s)$$

$$\boldsymbol{b_n = \frac{4}{T}\int_0^{T/2} f(t) \sin n\omega_0 t \, dt}$$



- Đồ thị đối xứng qua gốc tọa độ

Sum and Scalar Multiple

The Fourier coefficients of a sum $f_1 + f_2$ are the sums of the corresponding Fourier coefficients of f_1 and f_2.

The Fourier coefficients of cf are c times the corresponding Fourier coefficients of f.

EXAMPLE

Find the Fourier series of the function (Fig. 268)

$$f(x) = x + \pi \quad \text{if} \quad -\pi < x < \pi \qquad \text{and} \qquad f(x + 2\pi) = f(x).$$

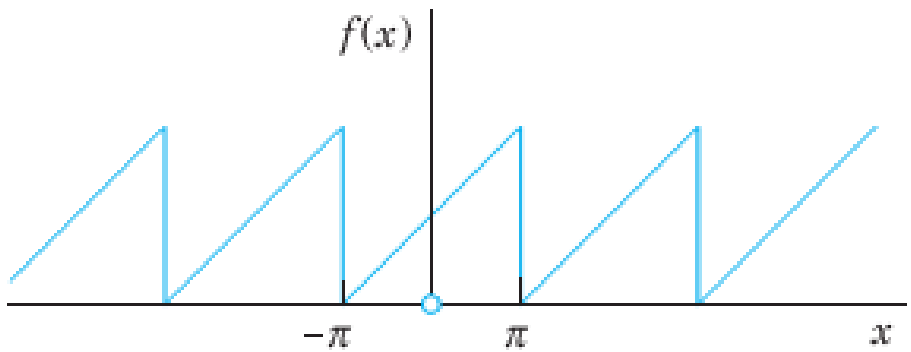


Fig. 268. The function $f(x)$. Sawtooth wave

Solution. We have $f = f_1 + f_2$, where $f_1 = x$ and $f_2 = \pi$. The Fourier coefficients of f_2 are zero, except for the first one (the constant term), which is π. Hence, by Theorem 1, the Fourier coefficients a_n, b_n are those of f_1, except for a_0, which is π. Since f_1 is odd, $a_n = 0$ for $n = 1, 2, \cdots$, and

$$b_n = \frac{2}{\pi}\int_0^{\pi} f_1(x)\sin nx\,dx = \frac{2}{\pi}\int_0^{\pi} x\sin nx\,dx.$$

Integrating by parts, we obtain

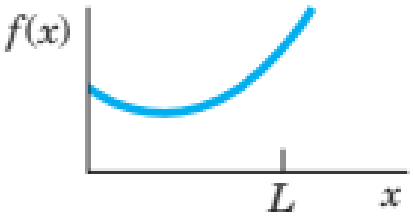
$$b_n = \frac{2}{\pi}\left[\frac{-x\cos nx}{n}\bigg|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos nx\,dx\right] = -\frac{2}{n}\cos n\pi.$$

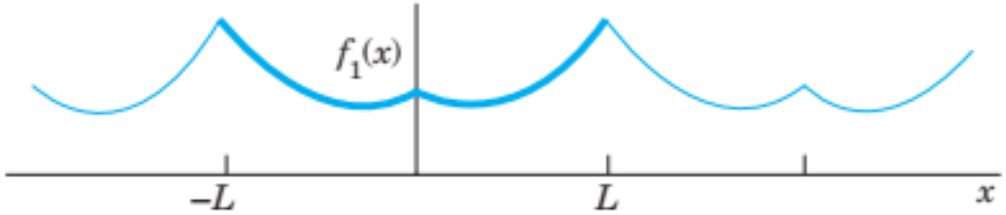
Hence $b_1 = 2$, $b_2 = -\frac{2}{2}$, $b_3 = \frac{2}{3}$, $b_4 = -\frac{2}{4}, \cdots$, and the Fourier series of $f(x)$ is

$$f(x) = \pi + 2\left(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \ + \cdots\right).$$

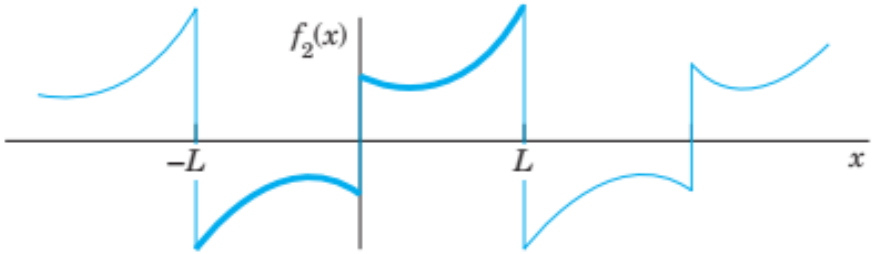
Half-Range Expansions

Hàm $f(t)$ có thể được biểu diễn theo chuỗi fourier bằng cách mở rộng hàm $f(t)$ thêm khoảng L như một hàm tuần hoàn với chu kỳ $2L$





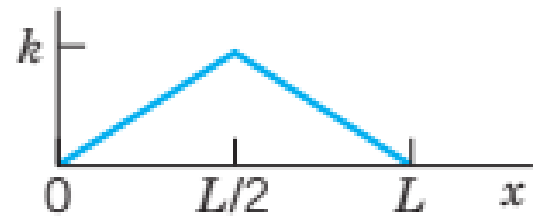
(a) $f(x)$ continued as an ***even*** periodic function of period $2L$



(b) $f(x)$ continued as an ***odd*** periodic function of period $2L$

EXAMPLE Find the two half-range expansions of the function

$$f(x) = \begin{cases} \dfrac{2k}{L}x & \text{if} \quad 0 < x < \dfrac{L}{2} \\[2ex] \dfrac{2k}{L}(L - x) & \text{if} \quad \dfrac{L}{2} < x < L. \end{cases}$$



Solution. **(a)** ***Even periodic extension.***

$$a_0 = \frac{1}{L}\left[\frac{2k}{L}\int_0^{L/2} x\,dx + \frac{2k}{L}\int_{L/2}^{L}(L - x)\,dx\right] = \frac{k}{2},$$

$$a_n = \frac{4k}{n^2\pi^2}\left(2\cos\frac{n\pi}{2} - \cos n\pi - 1\right).$$

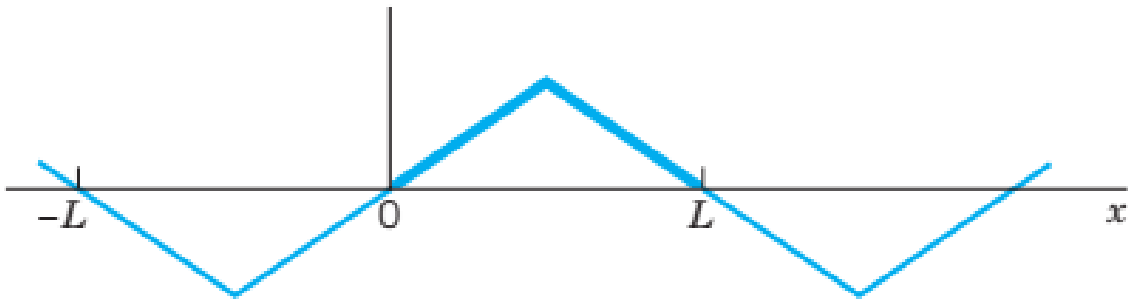
(a) Even extension

$$a_2 = -16k/(2^2\pi^2), \qquad a_6 = -16k/(6^2\pi^2), \qquad a_{10} = -16k/(10^2\pi^2), \cdots$$

$$f(x) = \frac{k}{2} - \frac{16k}{\pi^2}\left(\frac{1}{2^2}\cos\frac{2\pi}{L}x + \frac{1}{6^2}\cos\frac{6\pi}{L}x + \cdots\right).$$

Fourier cosine series.

(b) ***Odd periodic extension.***



(b) Odd extension

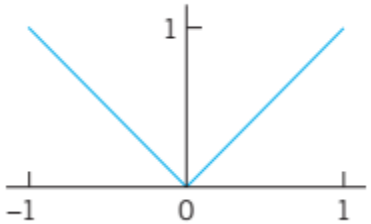
$$b_n = \frac{8k}{n^2\pi^2} \sin \frac{n\pi}{2}.$$

$$f(x) = \frac{8k}{\pi^2}\left(\frac{1}{1^2}\sin\frac{\pi}{L}x - \frac{1}{3^2}\sin\frac{3\pi}{L}x + \frac{1}{5^2}\sin\frac{5\pi}{L}x - + \cdots\right).$$ Fourier sine series.

PROBLEM SET 11.2

Find the Fourier series of the given function $f(x)$

8.



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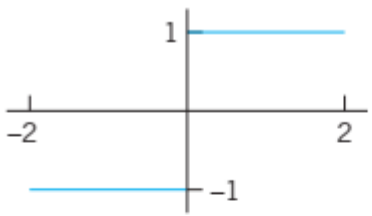

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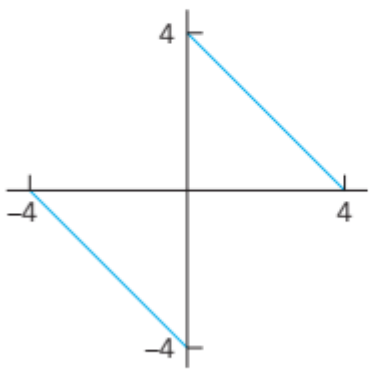
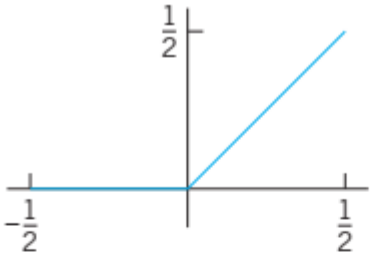


11. $f(x) = x^2 \quad (-1 < x < 1), \quad p = 2$

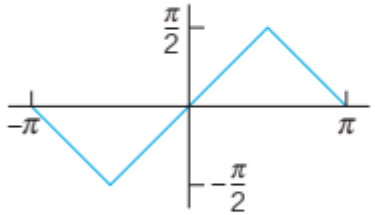
12. $f(x) = 1 - x^2/4 \quad (-2 < x < 2), \quad p = 4$

13.



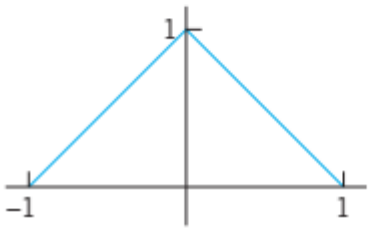
14. $f(x) = \cos \pi x \quad (-\frac{1}{2} < x < \frac{1}{2}), \quad p = 1$

15.



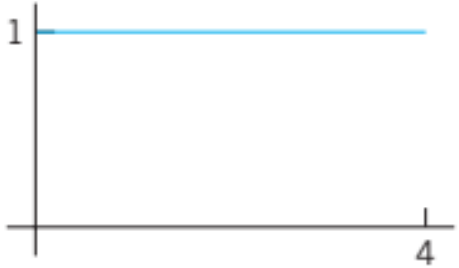
16. $f(x) = x|x| \quad (-1 < x < 1), \quad p = 2$

17.



Find (**a**) the Fourier cosine series, (**b**) the Fourier sine series.

23.



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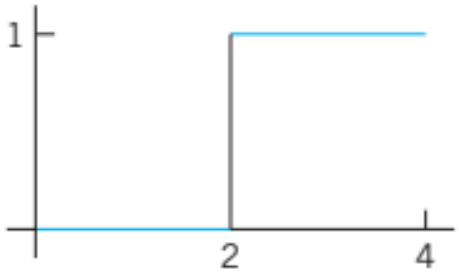

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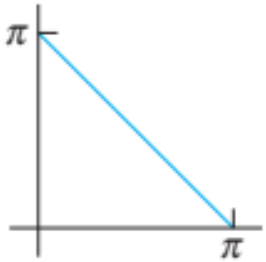

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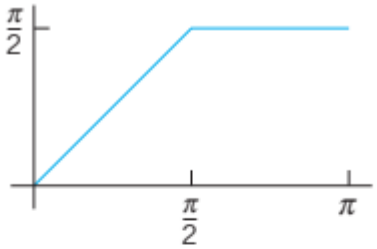

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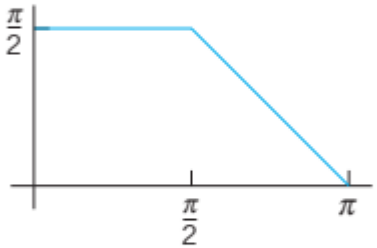

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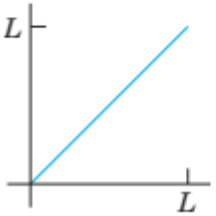
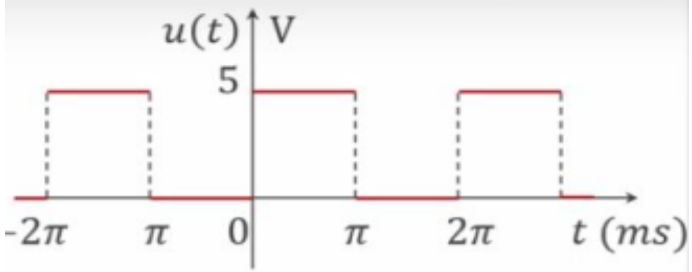
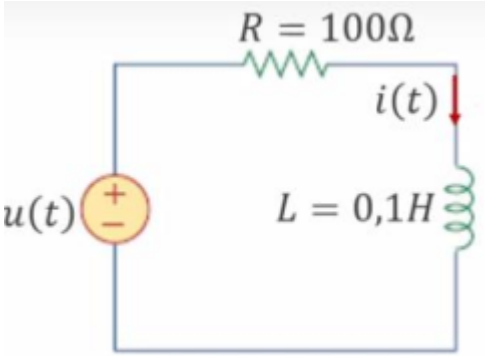


29. $f(x) = \sin x \ (0 < x < \pi)$

Ví dụ

tìm $i(t)$ nếu $n \leq 3$.

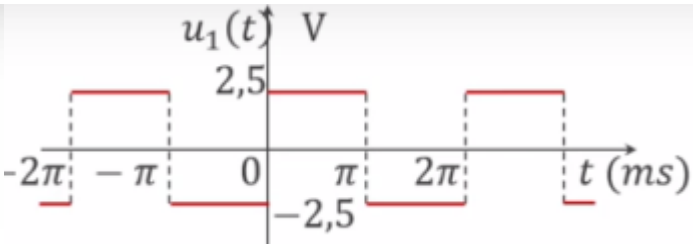




Giải

$$u(t) = u_1(t) + 2{,}5$$

$$T = 2\pi(ms) = 2\pi \times 10^{-3}(s); \quad \omega_0 = \frac{2\pi}{T} = 10^3(rad/s)$$



$$u_1(t) = \begin{cases} -2{,}5\ \text{V} & \text{nếu} -\pi < t < 0 \\ 2{,}5\ \text{V} & \text{nếu}\ 0 < t < \pi \end{cases}$$

Vì $u_1(t)$ là hàm lẻ nên $a_0 = 0$; $a_n = 0$

$$b_n = \frac{4}{T}\int_0^{T/2} 2{,}5 \sin n\frac{2\pi}{T}t\,dt = -\frac{4}{T}\frac{2{,}5}{n\frac{2\pi}{T}}\cos n\frac{2\pi}{T}t\Big|_0^{T/2} = -\frac{5}{n\pi}(\cos n\pi - \cos 0) = \begin{cases} \frac{10}{n\pi} & \text{nếu } n \text{ lẻ} \\ 0 & \text{nếu } n \text{ chẵn} \end{cases}$$

Với $n \leq 3$ thì

$$u_1(t) = \frac{10}{\pi}\sin 10^3 t + \frac{10}{3\pi}\sin 3 \times 10^3 t\ \text{V}$$

$$100i + 0{,}1i' = u(t) = 2{,}5 + \frac{10}{\pi}\sin 10^3 t + \frac{10}{3\pi}\sin 3\times 10^3 t \text{ V}$$

- Xét nguồn dc $u_{dc} = 2{,}5\text{V}$ tác động

$$100i_{dc} + 0{,}1{i_{dc}}' = 2{,}5 \Leftrightarrow 100i_{dc} = 2{,}5\text{V}$$

$$i_{dc} = 25\text{mA}$$

- Xét nguồn $u_1 = \frac{10}{\pi}\sin 10^3 t$ V tác động

$$100i_1 + 0{,}1{i_1}' = u_1 = \frac{10}{\pi}\sin 10^3 t$$

$$100I_1 + 0{,}1I_1 j10^3 = \frac{10}{\pi}\angle 0$$

$$I_1 = \frac{\frac{10}{\pi}}{100 + j100} = 22{,}5\angle -45^0\text{mA}$$

$$i_1 = 22{,}5\sin(10^3 t - 45^0)\text{ mA}$$

- Xét nguồn $u_3 = \frac{10}{3\pi}\sin(3\times 10^3 t)$ tác động

$$100i_3 + 0{,}1{i_3}' = u_3 = \frac{10}{3\pi}\sin(3\times 10^3 t)$$

$$100I_3 + 0{,}1I_3 j3\times 10^3 = \frac{10}{3\pi}\angle 0$$

$$I_3 = \frac{10}{3\pi}\times\frac{1}{100 + j300} = 3{,}35\angle -71{,}57^0\text{mA}$$

$$i_3 = 3{,}35\sin(3\times 10^3 t - 71{,}57^0)\text{ mA}$$

- Áp dụng định lý xếp chồng

$$i(t) = i_{dc} + i_1 + i_3 = 25 + 22{,}5\sin(10^3 t - 45^0) + 3{,}35\sin(3\times 10^3 t - 71{,}57^0)\text{mA}$$

Fourier integral

-Ideas and techniques of Fourier series extend to nonperiodic functions f(x)→the **Fourier integral**

(5)	$f(x) = \int_0^\infty [A(w)\cos wx + B(w)\sin wx]\, dw.$

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos wv\, dv, \qquad B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\sin wv\, dv$$

EXAMPLE

Find the Fourier integral representation of the function

$$f(x) = \begin{cases} 1 & \text{if} \quad |x| < 1 \\ 0 & \text{if} \quad |x| > 1 \end{cases}$$

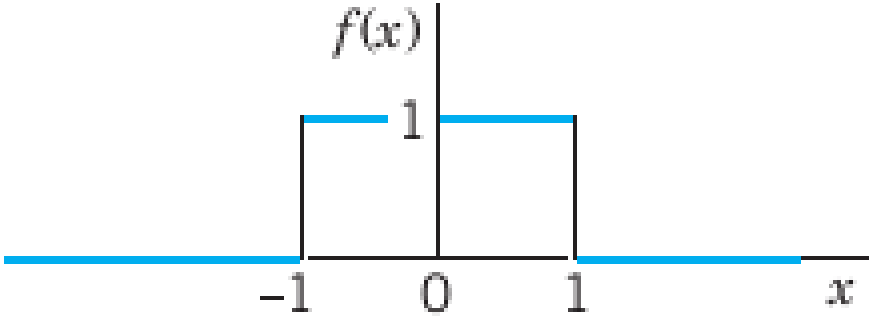


Fig. 281. Example 2

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Solution. From (4) we obtain

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(v)\cos wv\, dv = \frac{1}{\pi}\int_{-1}^{1}\cos wv\, dv = \frac{\sin wv}{\pi w}\bigg|_{-1}^{1} = \frac{2\sin w}{\pi w}$$

$$B(w) = \frac{1}{\pi}\int_{-1}^{1}\sin wv\, dv = 0$$

and (5) gives the *answer*

$$f(x) = \frac{2}{\pi}\int_{0}^{\infty}\frac{\cos wx \sin w}{w}\, dw. \tag{6}$$

Fourier cosine integral

- if f(x) has a Fourier integral representation and is even, then B(w)=0 and:

$$f(x) = \int_0^\infty A(w) \cos wx \, dw \tag{10}$$

where

$$A(w) = \frac{2}{\pi} \int_0^\infty f(v) \cos wv \, dv.$$

Fourier sine integral

- if f(x) has a Fourier integral representation and is odd, then A(w)=0 and:

$$f(x) = \int_0^\infty B(w) \sin wx \, dw \tag{11}$$

$$B(w) = \frac{2}{\pi} \int_0^\infty f(v) \sin wv \, dv.$$

Laplace Integrals

(13) $$\int_0^\infty \frac{\cos wx}{k^2 + w^2}\, dw = \frac{\pi}{2k} e^{-kx} \qquad (x > 0, \quad k > 0).$$

(15) $$\int_0^\infty \frac{w \sin wx}{k^2 + w^2}\, dw = \frac{\pi}{2} e^{-kx} \qquad (x > 0, \quad k > 0).$$

The integrals (13) and (15) are called the **Laplace integrals**.

PROBLEM SET 11.7

7–12 FOURIER COSINE INTEGRAL REPRESENTATIONS

Represent $f(x)$ as an integral (10).

7. $f(x) = \begin{cases} 1 & \text{if} \quad 0 < x < 1 \\ 0 & \text{if} \quad x > 1 \end{cases}$

8. $f(x) = \begin{cases} x^2 & \text{if} \quad 0 < x < 1 \\ 0 & \text{if} \quad x > 1 \end{cases}$

9. $f(x) = 1/(1 + x^2)$ $[x > 0.$ *Hint.* See (13).$]$

10. $f(x) = \begin{cases} a^2 - x^2 & \text{if} \quad 0 < x < a \\ 0 & \text{if} \quad x > a \end{cases}$

11. $f(x) = \begin{cases} \sin x & \text{if} \quad 0 < x < \pi \\ 0 & \text{if} \quad x > \pi \end{cases}$

12. $f(x) = \begin{cases} e^{-x} & \text{if} \quad 0 < x < a \\ 0 & \text{if} \quad x > a \end{cases}$

16–20 FOURIER SINE INTEGRAL REPRESENTATIONS

Represent $f(x)$ as an integral (11).

16. $f(x) = \begin{cases} x & \text{if} \quad 0 < x < a \\ 0 & \text{if} \quad x > a \end{cases}$

17. $f(x) = \begin{cases} 1 & \text{if} \quad 0 < x < 1 \\ 0 & \text{if} \quad x > 1 \end{cases}$

18. $f(x) = \begin{cases} \cos x & \text{if} \quad 0 < x < \pi \\ 0 & \text{if} \quad x > \pi \end{cases}$

19. $f(x) = \begin{cases} e^{x} & \text{if} \quad 0 < x < 1 \\ 0 & \text{if} \quad x > 1 \end{cases}$

20. $f(x) = \begin{cases} e^{-x} & \text{if} \quad 0 < x < 1 \\ 0 & \text{if} \quad x > 1 \end{cases}$

Fourier transform

An integral transform is a transformation in the form of an integral that produces from given functions new functions depending on a different variable

- Fourier Cosine Transforms
- Fourier Sine Transforms

Fourier Cosine Transforms

- The Fourier cosine transform concerns even functions f(x) and is obtained from the Fourier cosine integral as

$$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx$$

The inverse Fourier cosine transform

$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_c(w) \cos wx \, dw.$$

Fourier sine Transforms

- The Fourier sine transform concerns odd functions f(x) and is obtained from the Fourier sine integral as

$$\hat{f}_s(w) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \sin wx \, dx,$$

The inverse Fourier sine transform

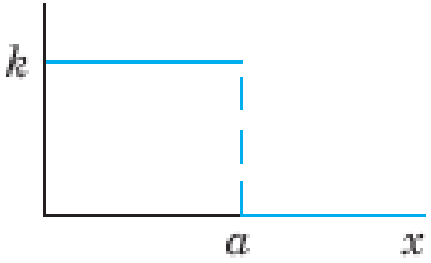
$$f(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty \hat{f}_s(w) \sin wx \, dw.$$

Other notations are

$$\mathcal{F}_c(f) = \hat{f}_c, \qquad \mathcal{F}_s(f) = \hat{f}_s$$

and $\mathcal{F}_c^{-1}$ and $\mathcal{F}_s^{-1}$ for the inverses of $\mathcal{F}_c$ and $\mathcal{F}_s$, respectively.

EXAMPLE



Find the Fourier cosine and Fourier sine transforms of the function

$$f(x) = \begin{cases} k & \text{if} \quad 0 < x < a \\ 0 & \text{if} \quad x > a \end{cases}$$

Solution. From the definitions (1a) and (2a) we obtain by integration

$$\hat{f}_c(w) = \sqrt{\frac{2}{\pi}}\, k \int_0^a \cos wx\, dx = \sqrt{\frac{2}{\pi}}\, k \left(\frac{\sin aw}{w} \right)$$

$$\hat{f}_s(w) = \sqrt{\frac{2}{\pi}}\, k \int_0^a \sin wx\, dx = \sqrt{\frac{2}{\pi}}\, k \left(\frac{1 - \cos aw}{w} \right).$$

EXAMPLE

Find $\mathscr{F}_c(e^{-x})$.

Solution. By integration by parts and recursion,

$$\mathscr{F}_c(e^{-x}) = \sqrt{\frac{2}{\pi}} \int_0^\infty e^{-x} \cos wx \, dx = \sqrt{\frac{2}{\pi}} \frac{e^{-x}}{1 + w^2} (-\cos wx + w \sin wx) \Big|_0^\infty = \frac{\sqrt{2/\pi}}{1 + w^2}.$$

Linearity

$$\boxed{\begin{aligned} &\text{(a)} \quad \mathcal{F}_c(af + bg) = a\mathcal{F}_c(f) + b\mathcal{F}_c(g), \\ &\text{(b)} \quad \mathcal{F}_s(af + bg) = a\mathcal{F}_s(f) + b\mathcal{F}_s(g). \end{aligned}}$$

Proof:

$$\begin{aligned} \mathcal{F}_c(af + bg) &= \sqrt{\frac{2}{\pi}} \int_0^\infty [af(x) + bg(x)] \cos wx \, dx \\ &= a\sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos wx \, dx + b\sqrt{\frac{2}{\pi}} \int_0^\infty g(x) \cos wx \, dx. \end{aligned}$$

Cosine and Sine Transforms of Derivatives

Let $f(x)$ be continuous and absolutely integrable on the x-axis, let $f'(x)$ be piecewise continuous on every finite interval, and let $f(x) \to 0$ as $x \to \infty$. Then

(4)

(a) $$\mathcal{F}_c\{f'(x)\} = w\mathcal{F}_s\{f(x)\} - \sqrt{\frac{2}{\pi}} f(0),$$

(b) $$\mathcal{F}_s\{f'(x)\} = -w\mathcal{F}_c\{f(x)\}.$$

$$\mathcal{F}_c\{f''(x)\} = -w^2\mathcal{F}_c\{f(x)\} - \sqrt{\frac{2}{\pi}} f'(0).$$

$$\mathcal{F}_s\{f''(x)\} = -w^2\mathcal{F}_s\{f(x)\} + \sqrt{\frac{2}{\pi}} wf(0).$$

Find the Fourier cosine transform $\mathcal{F}_c(e^{-ax})$ of $f(x) = e^{-ax}$, where $a > 0$.

Solution. By differentiation, $(e^{-ax})'' = a^2 e^{-ax}$; thus

$$a^2 f(x) = f''(x).$$

From this, (5a), and the linearity (3a),

$$\begin{aligned} a^2 \mathcal{F}_c(f) &= \mathcal{F}_c(f'') \\ &= -w^2 \mathcal{F}_c(f) - \sqrt{\frac{2}{\pi}} f'(0) \\ &= -w^2 \mathcal{F}_c(f) + a\sqrt{\frac{2}{\pi}}. \end{aligned}$$

Hence

$$(a^2 + w^2)\mathcal{F}_c(f) = a\sqrt{2/\pi}.$$

Complex Form of the Fourier Integral (called as Fourier transform)

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-iwx}\, dx.$$

The inverse Fourier transform

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(w) e^{iwx}\, dw$$

Another notation for the Fourier transform is

$$\hat{f} = \mathcal{F}(f),$$

so that

$$f = \mathcal{F}^{-1}(\hat{f}).$$

EXAMPLE

Find the Fourier transform of $f(x) = 1$ if $|x| < 1$ and $f(x) = 0$ otherwise.

Solution. Using (6) and integrating, we obtain

$$\hat{f}(w) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-iwx}\, dx = \frac{1}{\sqrt{2\pi}} \cdot \frac{e^{-iwx}}{-iw}\bigg|_{-1}^{1} = \frac{1}{-iw\sqrt{2\pi}}(e^{-iw} - e^{iw}).$$

As in (3) we have $e^{iw} = \cos w + i \sin w$, $e^{-iw} = \cos w - i \sin w$, and by subtraction

$$e^{iw} - e^{-iw} = 2i \sin w.$$

Substituting this in the previous formula on the right, we see that i drops out and we obtain the answer

$$\hat{f}(w) = \sqrt{\frac{\pi}{2}}\, \frac{\sin w}{w}.$$

EXAMPLE

Find the Fourier transform $\mathcal{F}(e^{-ax})$ of $f(x) = e^{-ax}$ if $x > 0$ and $f(x) = 0$ if $x < 0$; here $a > 0$.

Solution. From the definition (6) we obtain by integration

$$\mathcal{F}(e^{-ax}) = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-ax} e^{-iwx}\, dx$$

$$= \frac{1}{\sqrt{2\pi}} \left. \frac{e^{-(a+iw)x}}{-(a+iw)} \right|_{x=0}^{\infty} = \frac{1}{\sqrt{2\pi}(a+iw)}.$$

Linearity of the Fourier Transform

$$\mathcal{F}(af + bg) = a\mathcal{F}(f) + b\mathcal{F}(g).$$

Fourier Transform of the Derivative

$$\mathcal{F}\{f'(x)\} = iw\mathcal{F}\{f(x)\}.$$

$$\mathcal{F}\{f''(x)\} = -w^2\mathcal{F}\{f(x)\}.$$

Find the Fourier transform of xe^{-x^2} from Table III, Sec 11.10.

Solution. We use (9). By formula 9 in Table III

$$\begin{aligned}
\mathcal{F}(xe^{-x^2}) &= \mathcal{F}\{-\tfrac{1}{2}(e^{-x^2})'\} \\
&= -\tfrac{1}{2}\mathcal{F}\{(e^{-x^2})'\} \\
&= -\tfrac{1}{2}iw\mathcal{F}(e^{-x^2}) \\
&= -\frac{1}{2}iw\frac{1}{\sqrt{2}}e^{-w^2/4} \\
&= -\frac{iw}{2\sqrt{2}}e^{-w^2/4}.
\end{aligned}$$

Convolution

(11)

$$h(x) = (f * g)(x) = \int_{-\infty}^{\infty} f(p)\, g(x-p)\, dp = \int_{-\infty}^{\infty} f(x-p)\, g(p)\, dp.$$

(12)

$$\mathcal{F}(f * g) = \sqrt{2\pi}\, \mathcal{F}(f)\, \mathcal{F}(g).$$

Discrete Fourier Transform (DFT)

- A function f(x) is given only in terms of values at finitely many points.
- The main application of such a “discrete Fourier analysis” concerns large amounts of equally spaced data, as they occur in telecommunication, time series analysis, and various simulation problems.

Discrete Fourier Transform (DFT)

discrete Fourier transform of the given signal $\mathbf{f} = [f_0 \quad \cdots \quad f_{N-1}]^T$ to be the vector $\hat{\mathbf{f}} = [\hat{f}_0 \quad \cdots \quad \hat{f}_{N-1}]$ with components

(18)
$$\hat{f}_n = Nc_n = \sum_{k=0}^{N-1} f_k e^{-inx_k}, \qquad f_k = f(x_k), \quad n = 0, \cdots, N-1.$$

$$c_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-inx_k} \qquad x_k = \frac{2\pi k}{N}, \qquad k = 0, 1, \cdots, N-1.$$

→This is the frequency spectrum of the signal

$$\hat{\mathbf{f}} = \mathbf{F}_N \mathbf{f}, \text{ where the } N \times N \textbf{ Fourier matrix } \mathbf{F}_N = [e_{nk}]$$

$$e_{nk} = e^{-inx_k} = e^{-2\pi ink/N} = w^{nk}, \qquad w = w_N = e^{-2\pi i/N},$$

Let $N = 4$ measurements (sample values) be given. Then $w = e^{-2\pi i/N} = e^{-\pi i/2} = -i$ and thus $w^{nk} = (-i)^{nk}$. Let the sample values be, say $\mathbf{f} = [0 \quad 1 \quad 4 \quad 9]^{\mathsf{T}}$. Then by (18) and (19),

$$\hat{\mathbf{f}} = \mathbf{F}_4\mathbf{f} = \begin{bmatrix} w^0 & w^0 & w^0 & w^0 \\ w^0 & w^1 & w^2 & w^3 \\ w^0 & w^2 & w^4 & w^6 \\ w^0 & w^3 & w^6 & w^9 \end{bmatrix} \mathbf{f} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 14 \\ -4 + 8i \\ -6 \\ -4 - 8i \end{bmatrix}. \tag{20}$$

PROBLEM SET 11.9

Find the Fourier transform of $f(x)$

2. $f(x) = \begin{cases} e^{2ix} & \text{if} \quad -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

3. $f(x) = \begin{cases} 1 & \text{if} \quad a < x < b \\ 0 & \text{otherwise} \end{cases}$

4. $f(x) = \begin{cases} e^{kx} & \text{if} \quad x < 0 \quad (k > 0) \\ 0 & \text{if} \quad x > 0 \end{cases}$

5. $f(x) = \begin{cases} e^{x} & \text{if} \quad -a < x < a \\ 0 & \text{otherwise} \end{cases}$

6. $f(x) = e^{-|x|} \quad (-\infty < x < \infty)$

7. $f(x) = \begin{cases} x & \text{if} \quad 0 < x < a \\ 0 & \text{otherwise} \end{cases}$

8. $f(x) = \begin{cases} xe^{-x} & \text{if} \quad -1 < x < 0 \\ 0 & \text{otherwise} \end{cases}$

9. $f(x) = \begin{cases} |x| & \text{if} \quad -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

10. $f(x) = \begin{cases} x & \text{if} \quad -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

11. $f(x) = \begin{cases} -1 & \text{if} \quad -1 < x < 0 \\ 1 & \text{if} \quad 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$

21. Find the transform (the frequency spectrum) of a general signal of two values $[f_1 \quad f_2]^{\mathsf{T}}$.

Tables of Transforms
→page 534,535,536

11.10 Tables of Transforms

Table I. Fourier Cosine Transforms

Table II. Fourier Sine Transforms

Table III. Fourier Transforms